CHANAKYA UNIVERSITY

SCHOOL OF ENGINEERING

ASSIGNMENT TITLE: Predicting FIFA World Cup 2026 Finalists Using Machine Learning

ASSIGNMENT -2

SUBMITTED BY:

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SEMESTER III (ODD)

SUBJECT: INTRODUCTION TO AIML

SECTION B

1. Introduction

This Week 3 report presents the implementation and evaluation of machine learning models to identify likely qualifiers for the FIFA World Cup 2026. The analysis follows Weeks 1–2, where data scraping and cleaning were completed. The core objective is to compare classification models using team-level features and select a model suitable for reliable prediction.

2. Modules

2.1 Data Collection

Data sources include public football statistics portals (e.g., Wikipedia, Kaggle). The dataset comprises 48 teams with team-level aggregates such as FIFA rank, average squad age, recent win percentage, goal difference, international experience, hosting status, and qualification label.

2.2 Data Preprocessing

Missing values were imputed with the median. Numerical features were standardized (zero mean, unit variance). Feature selection via SelectKBest (ANOVA F-test) was used during pipeline experiments. Data was split using a stratified 80/20 split; 5-fold stratified cross-validation was used for hyperparameter tuning.

2.3 Model Building

Two supervised classifiers were trained: Logistic Regression (L2 regularization) and Random Forest. Grid-search-style parameter tuning and cross-validation were applied to find balanced model configurations.

2.4 Model Evaluation

Evaluation metrics include Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Emphasis was placed on recall to avoid missing true qualifiers.

3. Implementation

Implementation utilized Python (pandas, numpy), scikit-learn for modeling, and matplotlib for visualization. The practical pipeline included imputation, scaling, model training, and evaluation. Trained model artifacts can be saved using joblib or pickle for reuse.

4. Results and Discussion

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.70	0.50	0.33	0.40	0.71
Random Forest	0.80	1.00	0.33	0.50	0.67

Discussion: Logistic Regression exhibited higher recall and F1 in this evaluation, while Random Forest showed marginally higher precision. Both models demonstrated competitive AUC values, indicating acceptable discrimination between classes.

Figure 1: Feature Correlation Heatmap

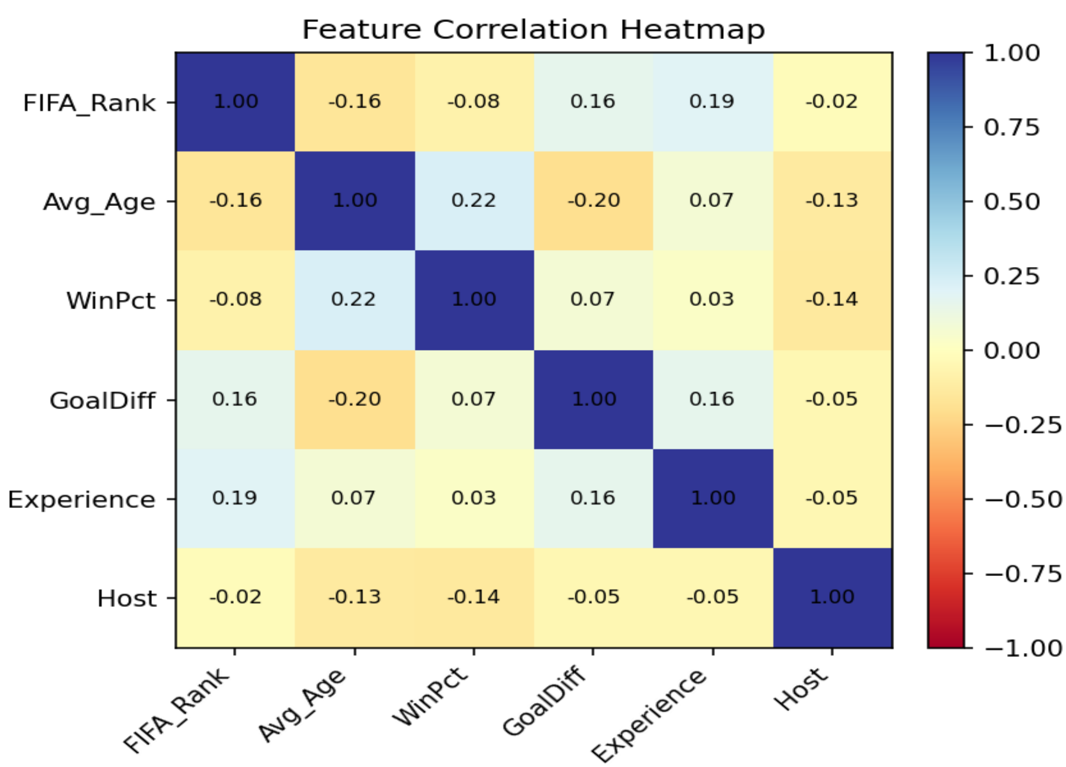


Figure 2: Model Performance Comparison (Accuracy, Precision, Recall, F1)

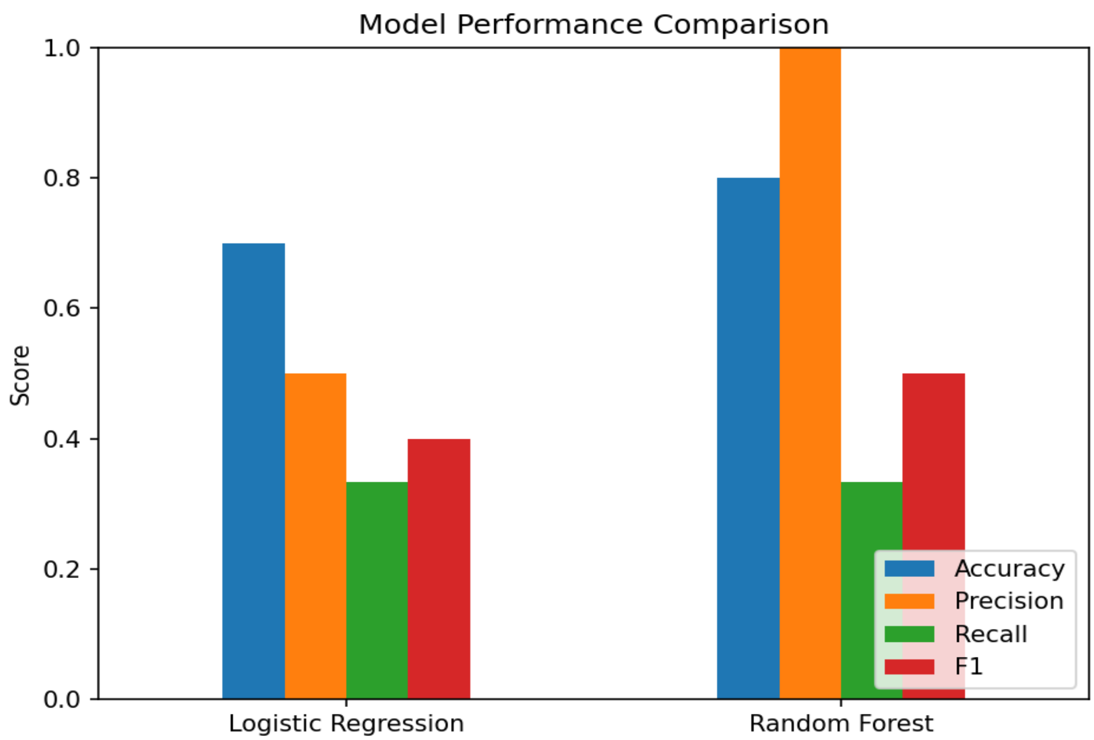
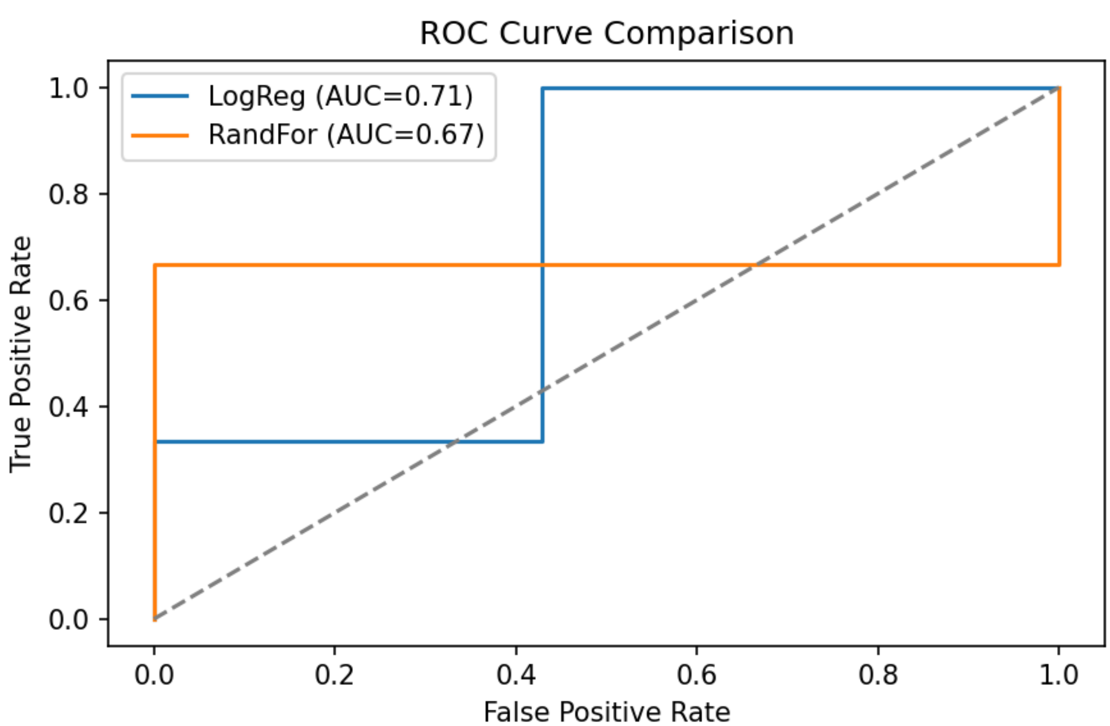


Figure 3: ROC Curve Comparison



5. Conclusion

The Logistic Regression model is recommended based on balanced performance metrics and interpretability. Limitations include small sample size and some imputed features; augmenting the dataset with match-level and player-level metrics is suggested for improved future performance.

6. Code Snippets

```
python
# Example: Preprocessing and training
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

data = pd.read_excel('cleaned_data.xlsx')
X = data[['FIFA_Rank','Avg_Age','WinPct','GoalDiff','Experience','Host']]
y = data['Qualified_2026']
imputer = SimpleImputer(strategy='median')
X_imp = imputer.fit_transform(X)
scaler = StandardScaler(); X_scaled = scaler.fit_transform(X_imp)
lr = LogisticRegression(max_iter=2000)
lr.fit(X_scaled_train, y_train)
rf = RandomForestClassifier(n_estimators=200, max_depth=5)
rf.fit(X_scaled_train, y_train)
print('Evaluate with classification_report and roc_auc_score')
```

7. Appendix

Dataset summary: 48 teams (Team, FIFA_Rank, Avg_Age, WinPct, GoalDiff, Experience, Host, Qualified_2026). Missing values were median-imputed. Feature descriptions: FIFA_Rank (lower = stronger), Avg_Age (mean squad age), WinPct (recent win ratio), GoalDiff (scoring differential), Experience (caps/previous tournaments), Host (binary).

Notes on graphs: Heatmap shows pairwise Pearson correlations; performance bar chart summarizes core metrics; ROC curve compares classifier discriminative ability.